StudyPilot: Adaptive Practice Engine for Math and Science

Concept Note | IBM SkillsBuild Masterclass 5: Creating Real Life Projects

1. Problem Statement

Students usually practise from the same textbook exercises or the same homework sheet, whatever they have already mastered. Strong topics get over-practised while weak ones are missed. Teachers have little time to review quiz results topic by topic for every student, so gaps are often found only at exam time, when it is too late to fix them calmly.

2. Proposed Solution

StudyPilot is an adaptive learning engine, similar in spirit to tools such as Carnegie Learning's MATHia. It does three things:

- **Tracks quiz history.** Every answer is logged by student, topic, difficulty and time taken.
- **Identifies weak topics.** A mastery model estimates how well the student knows each math or science topic and flags the weakest.
- **Serves customised practice.** The recommender chooses the next problems from the question bank, matching the weak topics and a difficulty that is challenging but achievable.

Students see their progress per topic, and teachers see weak topics across the class.

3. Target Users

- **Primary:** students in grades 8 to 12 in India practising math and science, especially Class 9 and 10 students.
- **Secondary:** teachers, tutors, coaching centres and parents who want topic-level visibility.

4. How It Works

1. The student takes a short diagnostic quiz, then regular quizzes.
2. Each answer is logged with its topic, difficulty and whether it was correct.
3. The mastery model updates a score for every topic, starting with a simple rule-based score and moving to Bayesian Knowledge Tracing.
4. The recommender picks the next set of problems from the weakest topics at a suitable difficulty.
5. The student and teacher dashboards show progress, and new answers keep updating the model.

5. Unique Value

Textbooks and generic quiz apps give everyone the same questions. Private tutors personalise well but cost more. StudyPilot gives each student a tutor-like path through the material at low cost, with feedback after each answer and a topic-by-topic view for teachers.

6. Business Model

- Free tier with limited daily practice.
- Premium student plan with unlimited practice, detailed reports and revision plans.
- School licences with a teacher dashboard, class analytics and curriculum-aligned question banks.

7. Expected Impact

- Students spend practice time on the topics they find hardest.
- Teachers see class-wide weak topics without marking every quiz by hand.
- Gaps are found early, not at exam time.
- Students build steady study habits through short, targeted practice.

8. Success Metrics

- Quiz score improvement between a student's first and latest quizzes.
- Weekly active learners and retention.
- Accuracy of the model's predictions of whether a student will answer correctly.
- Free-to-paid conversion and number of schools onboarded.

9. Risks and Mitigation

- **New students with no history:** start with a short diagnostic quiz so the first recommendations are useful.
- **Question quality:** use a teacher-reviewed question bank tagged by topic and difficulty.
- **Student privacy:** collect only what is needed, keep data encrypted, and get school and parent consent for minors.
- **Frustration or boredom:** mix in easier problems and review items so practice stays motivating.
- **Inaccurate recommendations:** keep the teacher in the loop and let students give feedback on problems.

10. Next Steps

- Build a prototype for one subject: a tagged question bank, quiz logging and a rule-based mastery score.
- Add the recommender and a simple student dashboard, then test with a small group of students and one teacher.
- Improve the model with the collected data, then pilot with one school.